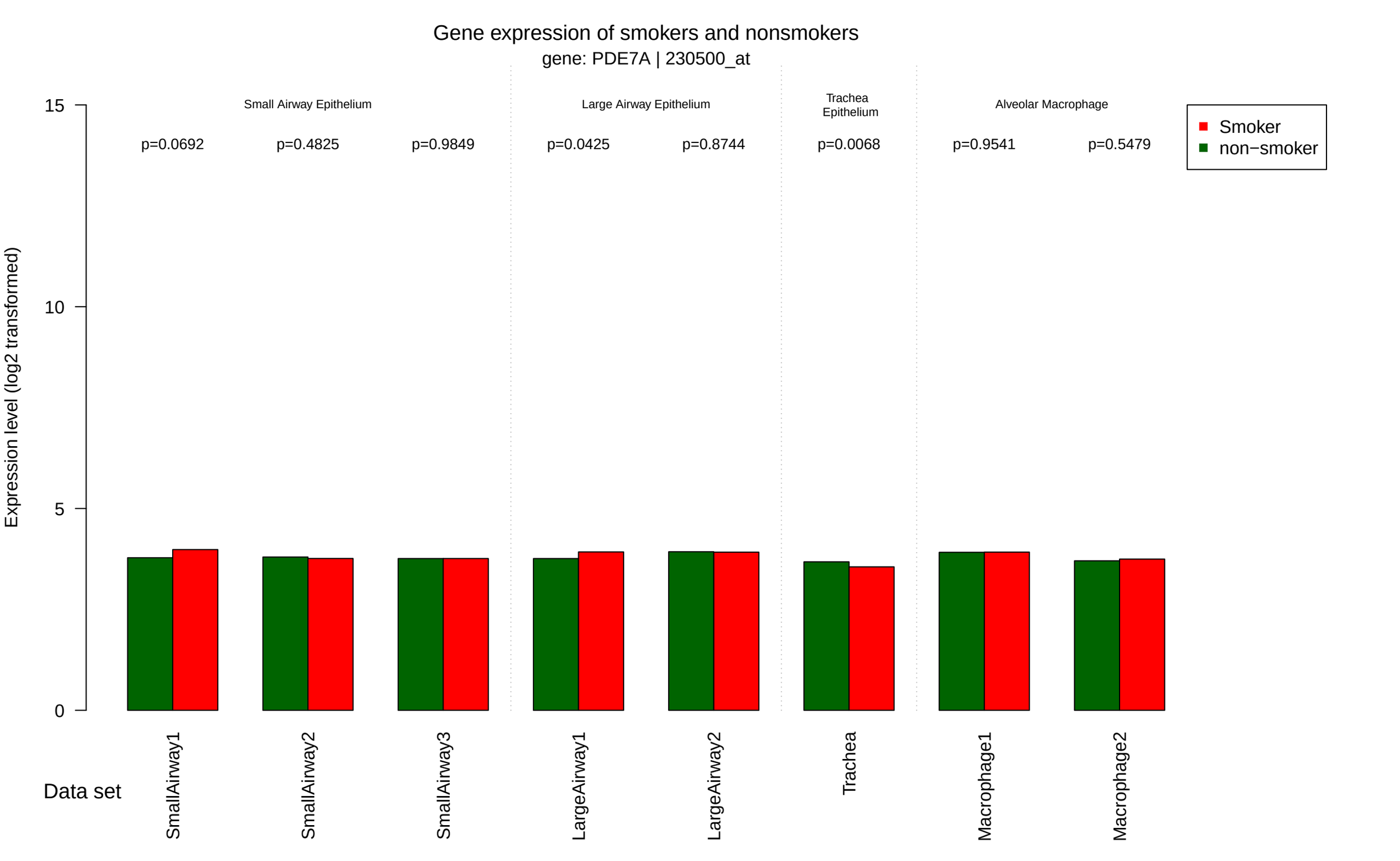
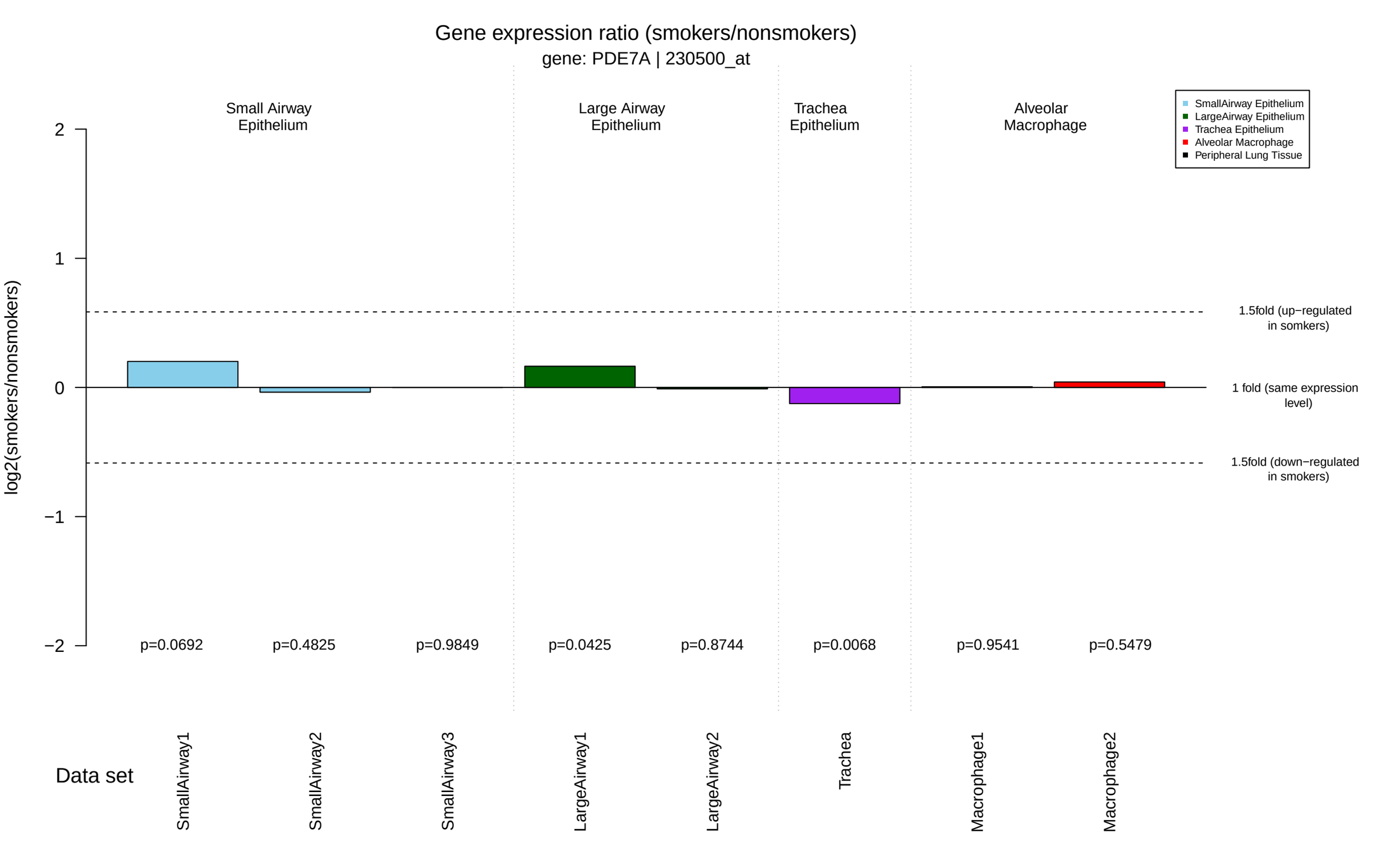
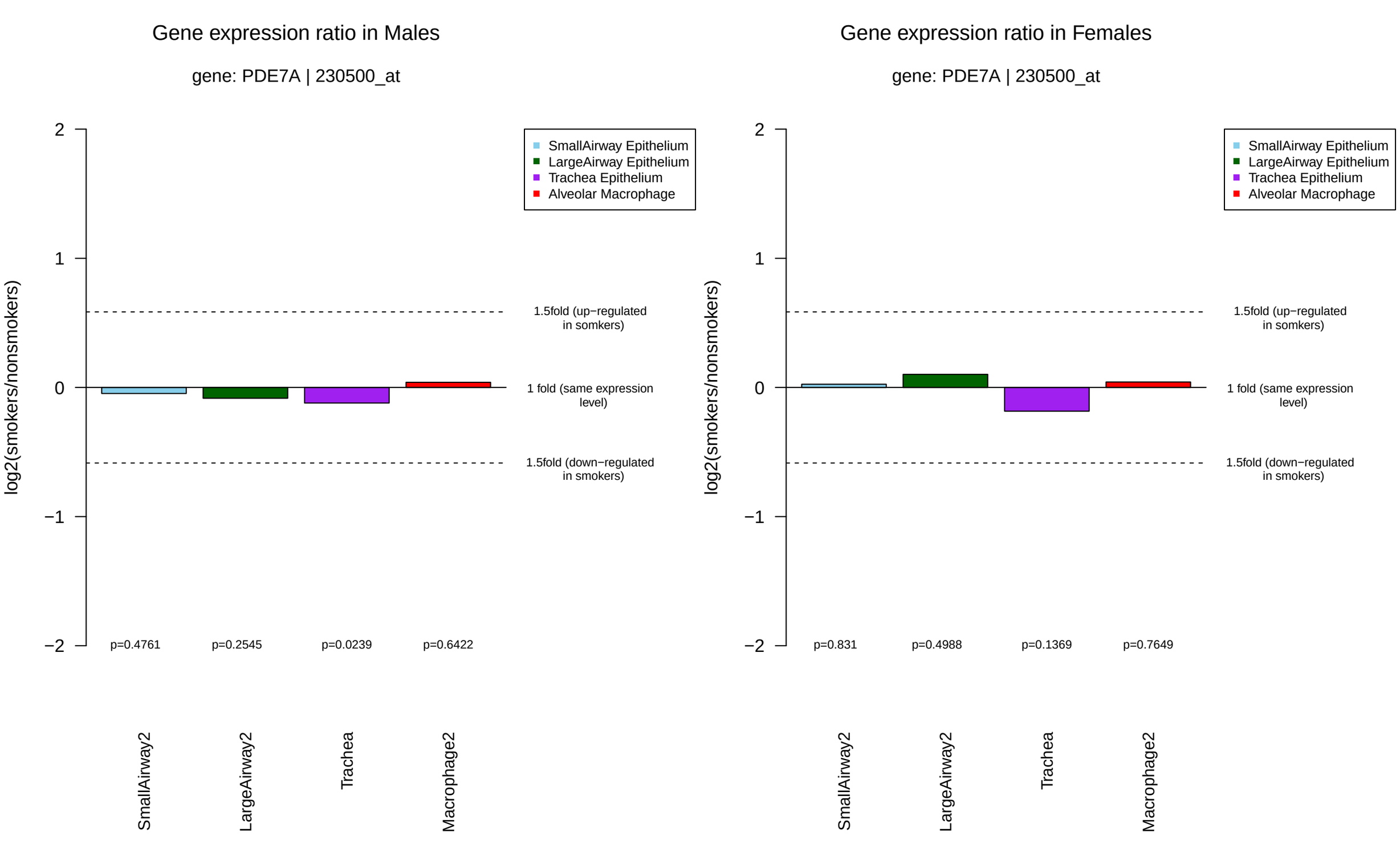
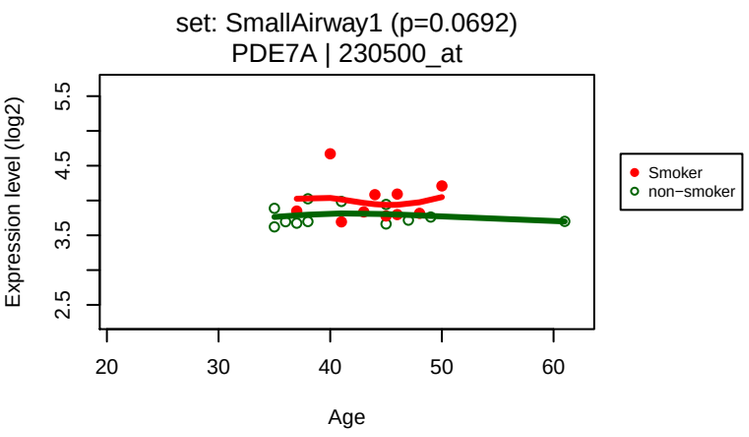
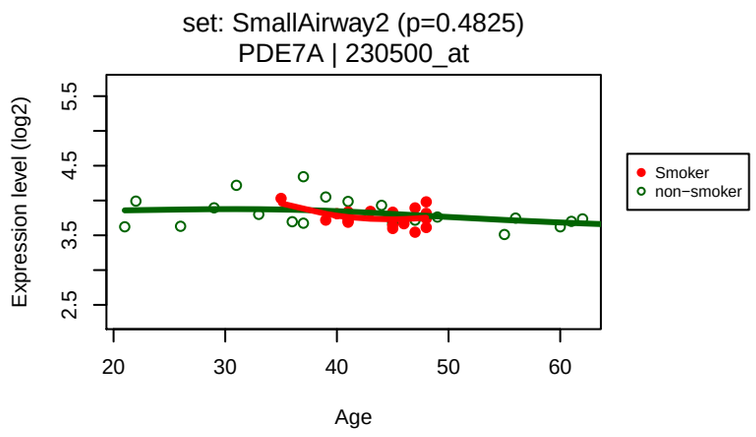
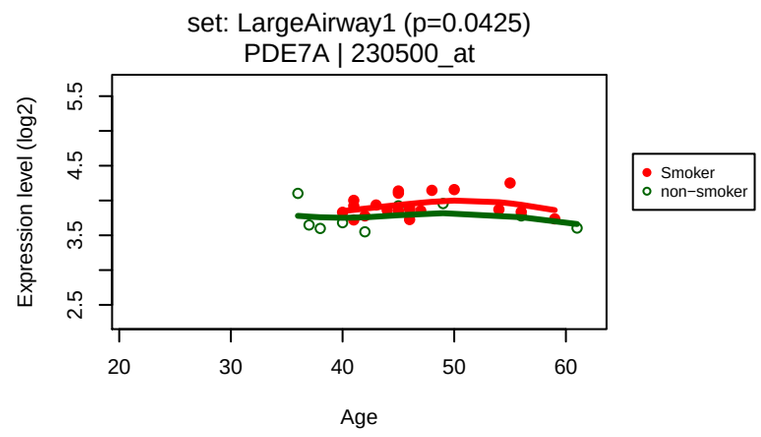

Gene expression of smokers and nonsmokers
gene: PDE7A | 230500_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0692
p=0.4825
p=0.9849
p=0.0425
p=0.8744
p=0.0068
p=0.9541
p=0.5479
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

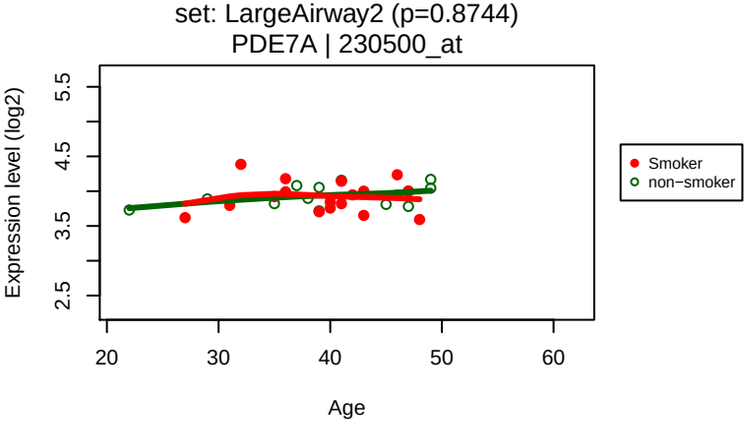

Gene expression ratio (smokers/nonsmokers)
gene: PDE7A | 230500_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0692
p=0.4825
p=0.9849
p=0.0425
p=0.8744
p=0.0068
p=0.9541
p=0.5479
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

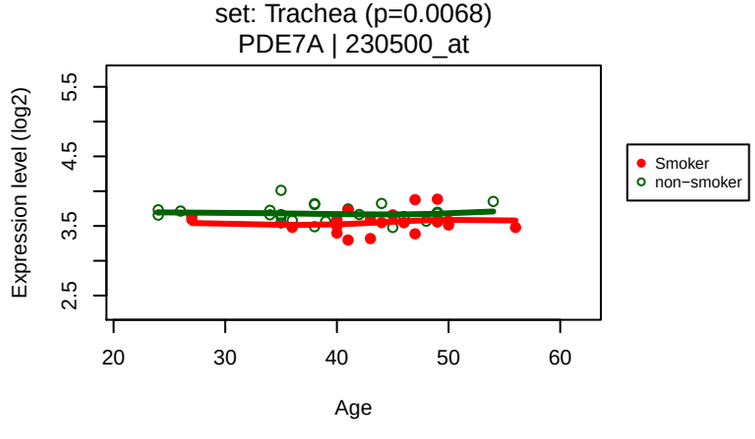

Gene expression ratio in Males
gene: PDE7A | 230500_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4761
p=0.2545
p=0.0239
p=0.6422
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE7A | 230500_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.831
p=0.4988
p=0.1369
p=0.7649
SmallAirway2
LargeAirway2
Trachea
Macrophage2

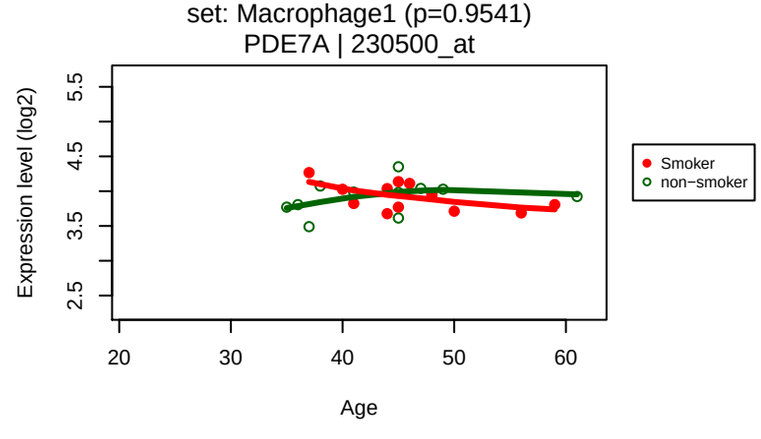

set: SmallAirway1 (p=0.0692)
PDE7A | 230500_at
Expression level (log2)
Age
Smoker
non−smoker

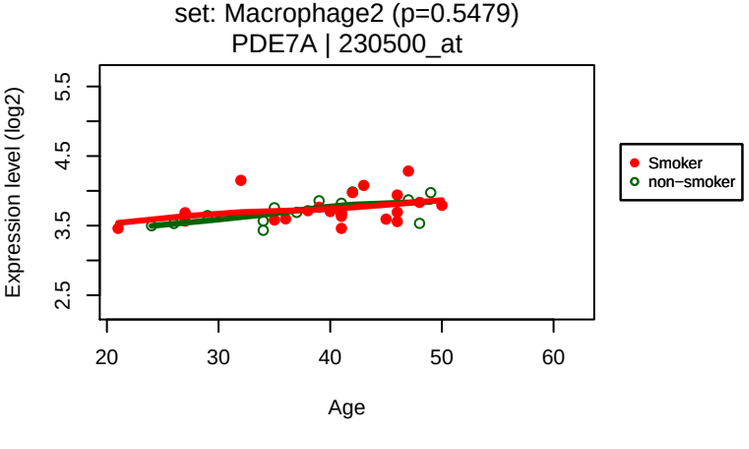

set: SmallAirway2 (p=0.4825)
PDE7A | 230500_at
Expression level (log2)
Age
Smoker
non−smoker

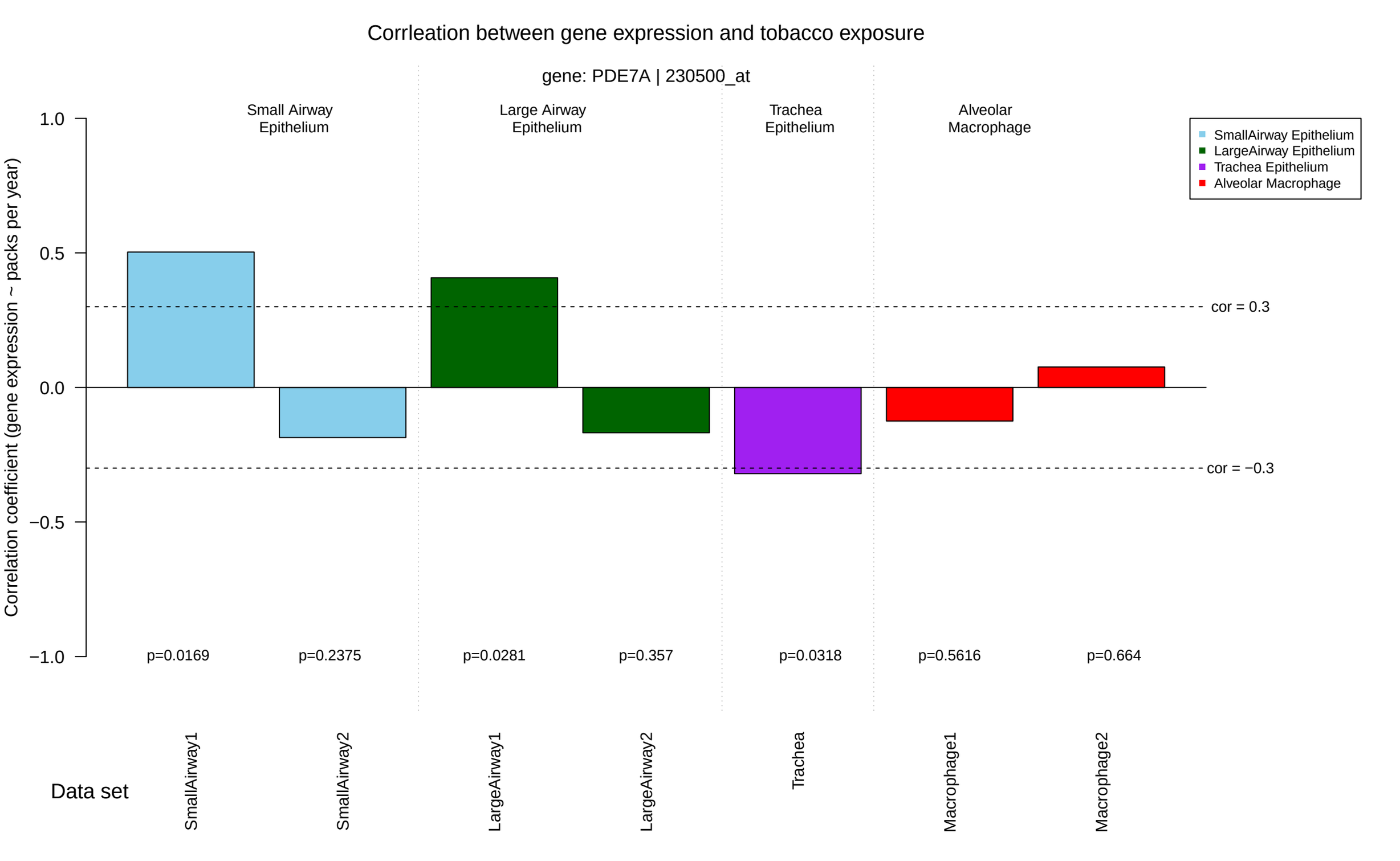

set: LargeAirway1 (p=0.0425)
PDE7A | 230500_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8744)
PDE7A | 230500_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0068)
PDE7A | 230500_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9541)
PDE7A | 230500_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5479)
PDE7A | 230500_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE7A | 230500_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0169
p=0.2375
p=0.0281
p=0.357
p=0.0318
p=0.5616
p=0.664
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2